# Kavya Agar

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## EDUCATION

**Texas A&M University** **May 2028**
*Bachelor of Science in Computer Science — Minor in Business, Minor in Statistics* *GPA: 3.5*
– **Coursework:** Data Structures & Algorithms, Computer Systems, Machine Learning, Linear Algebra, Probability & Statistics
– **Organizations:** Texas A&M Computing Society, Product@TAMU, SASE, Aggie Data Science Club

## SKILLS

**Languages & Frameworks:** Python, Java, C/C++, JavaScript/TypeScript, SQL, React, FastAPI, Django, PostgreSQL
**Cloud & AI/ML:** AWS (EC2, S3, Lambda, Step Functions, SageMaker), Docker, PyTorch, Hugging Face, Anthropic SDK, MCP, Claude Code
**Product & Tools:** Figma, Jira, Salesforce, Agile/Scrum, Git, CI/CD

## PROFESSIONAL EXPERIENCE

**Aspen Technology (an Emerson company) – Business Development Intern** **Houston, TX**
*Sales & Business Development* *May 2026 – Aug 2026*
– Designed and built an internal CRM-style account intelligence platform covering **400+ priority accounts**, mapping decision-maker personas via ZoomInfo and Industrial Info Resources to streamline prospecting for a **5-person BDR team**
– Conducted global market analysis across **40,000+ accounts** in Metals & Mining and Power Generation, identifying **20,000+ new potential accounts** and presenting expansion opportunities to **AspenTech's President** and senior **Emerson** leadership
– Developed AI-assisted workflows using Claude to accelerate account research, persona identification, and market-data synthesis, significantly reducing manual research time

**Aspen Technology (an Emerson company) – Technical Business Analyst Intern** **Houston, TX**
*Analytics Platforms (Python, BeautifulSoup, Selenium, Excel)* *Jan 2024 – Apr 2024*
– Automated competitive analysis of **5+ analytics platforms** via Python/Selenium pipelines, cutting manual research time **60%** and shaping feature-prioritization decisions tied to an estimated **15% revenue impact**

**JPMorgan Chase & Co. – Software Engineering Intern** **Houston, TX**
*Team Lead (Java, Apache Cassandra, AWS S3, Parquet, Agile)* *Jun 2023 – Aug 2023*
– Engineered a multithreaded distributed Java pipeline migrating **10TB+** of Cassandra data to AWS S3 in Parquet, improving throughput **90%** while preserving fault tolerance and data integrity; partnered with 3 senior engineers across **10+ code reviews** on reliability standards
– Led daily Agile stand-ups for an **8-person cross-functional team**, surfacing blockers early and aligning sprint priorities to delivery timelines

## PROJECTS

***CommitHunter (Python, FastAPI, PostgreSQL, SciPy, scikit-learn)*** *May 2026 – Aug 2026*
– Built a service correlating production error spikes with commit and deploy history to rank likely culprit commits — **3.7K LOC** across ingest, detection, ranking, FastAPI, and CLI layers, backed by **165 pytest tests**
– Implemented two-stage regression detection: **PELT** change-point search (dependency-free **CUSUM** fallback) proposes candidates, then an exact binomial test with **Benjamini–Hochberg FDR control** confirms them; modeled **deploys, not commits**, as the unit of attribution in a 6-table Postgres schema

***Quant Options Engine (Python, FastAPI, React, SciPy, yfinance, Anthropic SDK, MCP)*** *May 2026 – Present*
– Derived Black-Scholes Greeks analytically and built an implied-volatility solver via **Brent's method**, validating Monte Carlo convergence (**50K simulations**) with antithetic variate variance reduction across live SPY options chains
– Built an **MCP server** and Claude-driven strategy planner (raw Anthropic SDK tool-use loop) converting natural-language trading goals into multi-leg options strategies with exact, tool-computed P&L and Greeks, gated by a deterministic risk verifier and **15-scenario eval harness**

***Financial NER Analyzer (Python, PyTorch, Hugging Face, AWS, React, Docker)*** *Aug 2025 – Nov 2025*
– Fine-tuned **DistilBERT** for custom NER on transaction datasets (**98% validation accuracy**) and architected a fault-tolerant AWS pipeline (**Step Functions, Textract, SageMaker**) processing **500+ documents/week**

***Product Strategy Writing (kavyaagar.com)*** *Apr 2026 – Present*
– Published **6 long-form product teardowns** (Robinhood, Stripe, Wise, Venmo, Klarna, Amex) decomposing **monetization strategy**, **behavioral design**, and **unit economics**; sourced FINRA enforcement actions, CFPB circulars, DOJ settlements, and EU consumer-credit directives to support original critiques
– Authored **3 product strategy essays** on dark patterns in credit products and the convergence of banking and software, citing FTC and CFPB research

## CERTIFICATIONS

**AWS Certified Cloud Practitioner** (CLF-C02), Jun 2025 | **IBM DevOps & Software Engineering**, Jul 2025 | **Databricks GenAI Fundamentals** | **BCG GenAI Job Simulation**